DIA·SPANISH

Text-to-Dialogue for Spanish Conversations

Adapting NARI Labs' 1.6B dialogue model to natural multi-speaker Spanish audio

144 h	43,219	80	6
curated audio	training chunks	episodes	podcasts

D8 LABS · TECHNICAL REPORT · JUNE 2026

1 Overview

Dia is a 1.6-billion-parameter **text-to-dialogue** model: from a transcript annotated with speaker tags (`[S1]`, `[S2]`) and nonverbal cues (`(risas)`, `(suspira)`), it generates a single, continuous multi-speaker audio stream with natural turn-taking, intonation and emotion. The public model is **English-only**. This project adapts it to **Spanish**.

The key insight is that **nothing in the architecture is English-specific**. Dia reads text as raw UTF-8 bytes — there is no tokenizer and no fixed vocabulary — so Spanish glyphs (ñ, á, é, ü) are already representable. Audio is modelled as discrete Descript Audio Codec (DAC) tokens, which are language-agnostic. The only missing ingredient is **Spanish training data**, and that is what the bulk of this work delivers: a clean, diarized 144-hour conversational corpus and an end-to-end fine-tuning pipeline on Modal.

What was built

- **A reproducible data pipeline** — collection → ASR + diarization → chunking → audio cutting → HuggingFace packaging.
- **A 144-hour Spanish dialogue corpus** — 43,219 speaker-tagged clips from 80 episodes across 6 podcasts, staged on a Modal volume.
- **A full fine-tuning harness** — the inference-only Dia code was made trainable (loss over DAC codebooks, delay-pattern targets, 8-bit optimizer).
- **Validated training + inference loops on Modal** — smoke-scale checkpoints and Spanish audio samples generated end-to-end.

Status at a glance

Stage	State	Detail
Data collection	**Done**	RSS / podcast scraping, 80 episodes
Transcription + diarization	**Done**	WhisperX, word-level alignment + speakers
Chunking + audio cutting	**Done**	43,219 clips @ 44.1 kHz on Modal volume
HF dataset packaging	**Done**	92-shard Arrow dataset on volume
Fine-tuning harness	**Done**	Full FT, bf16, 8-bit AdamW
Smoke training run	**Done**	1 epoch / ~1k steps, 3 checkpoints saved
Full training run	**Pending**	43k chunks × 2 epochs on A100-80GB
Evaluation (WER/CER/MOS)	**Pending**	ASR-based + speaker consistency

2 Model Architecture

Dia is an encoder–decoder transformer. The **encoder** consumes byte-level text; the **decoder** autoregressively generates DAC audio codes across 9 parallel codebook channels, conditioned on the encoder output via cross-attention. A **delay pattern** staggers the 9 channels in time so coarse codes are predicted before the fine codes that depend on them.

Component	Configuration
Text input	Raw UTF-8 bytes — vocab 256, no tokenizer (Spanish-ready by construction)
Encoder	12 layers · d=1024 · 16 heads · MLP 4096
Decoder	18 layers · d=2048 · MLP 8192 · GQA (16 query / 4 KV heads)
Audio codec	Descript Audio Codec (DAC) · 9 codebooks · target vocab 1028
Delay pattern	[0, 8, 9, 10, 11, 12, 13, 14, 15] across the 9 channels
Sequence limits	Text 512 · audio 1536 frames
Precision	bfloat16
Parameters	~1.6 B

Inference flow

Spanish transcript → UTF-8 bytes → **Encoder** (12L) → **Decoder** (18L, cross-attn) → 9-channel DAC codes (delay pattern) → **DAC decoder** → 44.1 kHz waveform

Why Spanish needs no surgery

Because text enters as bytes and audio is DAC tokens, the embedding tables, attention and codec are all language-neutral. Adapting to Spanish is therefore a pure **data + weight-update** problem: feed the model Spanish (text, audio) pairs and let gradient descent move the distribution. No vocabulary expansion, no architecture change, no re-training of the codec.

3 Data Pipeline

The corpus is produced by a six-stage pipeline. Stages run locally for collection/ASR and on **Modal** (serverless GPU) for the audio-heavy and training steps, with all artifacts living on the `dia-spanish-vol` volume.

Stage	Script	What it does
1 · Collect	`collect_via_rss.py`	Scrape public Spanish podcast feeds via RSS (bypasses YouTube bot-detection); download source audio + metadata.
2 · Transcribe	`transcribe_whisperx.py`	WhisperX ASR with word-level timestamps and speaker diarization; emit per-episode JSON.
3 · Chunk	`scripts/chunk_for_training.py`	Group diarized words into 3–17 s clips with [S1]/[S2] tags; write chunk_manifest.jsonl (text + spans).
4 · Cut	`cut_audio_from_manifest.py`	Slice source WAVs to per-chunk clips, resample 16k→44.1 kHz mono; write clips + _metadata.jsonl.
5 · Package	`build_hf_dataset.py`	Build a HuggingFace Dataset with an Audio feature; save 92-shard Arrow to the volume / Hub.
6 · Train	`train_dia_es.py + dia/finetune.py`	Modal job: DAC-encode audio, full fine-tune Dia, checkpoint to volume; infer_dia_es.py for sampling.

Chunk format

Each training example is one clip plus its transcript line, e.g.:

```
{ "chunk_id": "aladetres_042686b90050_0000", "start": 0.231, "end": 16.919,
  "duration": 16.69, "speakers": ["[S1]"], "num_turns": 1,
  "text": "[S1] La Segunda Guerra Mundial se produce porque Alemania rompe
  el equilibrio geopolítico en Europa de un modo muy veloz y agresivo…" }
```

4 Data Distribution

The corpus totals **144.4 hours** across **43,219 chunks** and 80 episodes from 6 conversational podcasts spanning Spain and Latin-American accents. Sources range from interview and talk formats (The Wild Project, Yordi Rosado) to narrative journalism (Radio Ambulante) and psychology / culture shows.

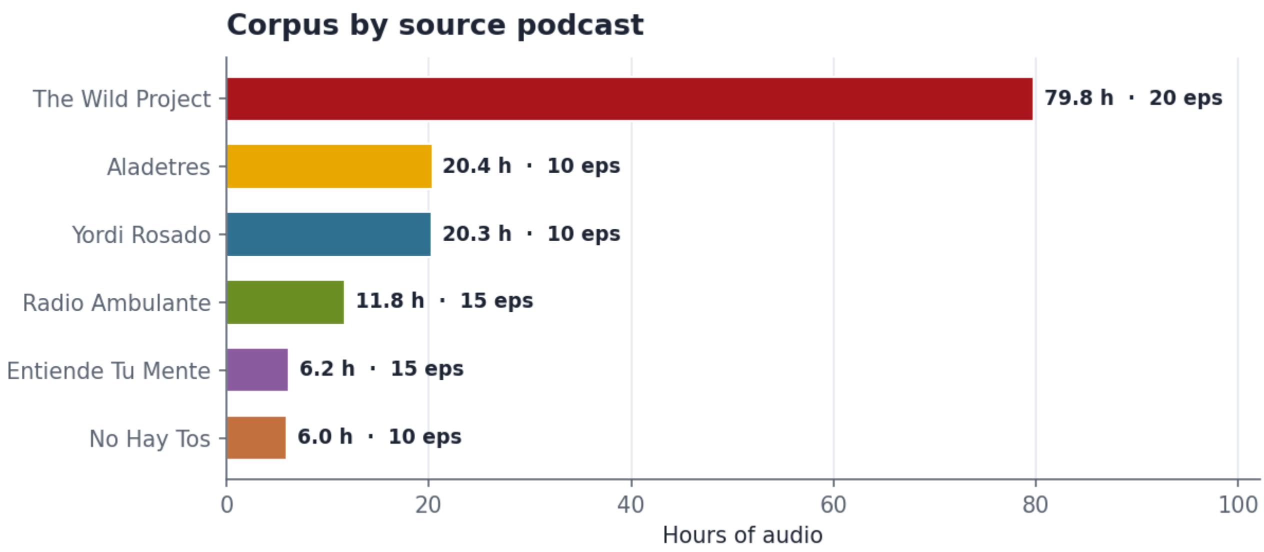

Figure 1 — Hours of audio and episode count per source. The Wild Project dominates (~55% of audio); a planned full run should weight sampling to avoid over-fitting to a single voice.

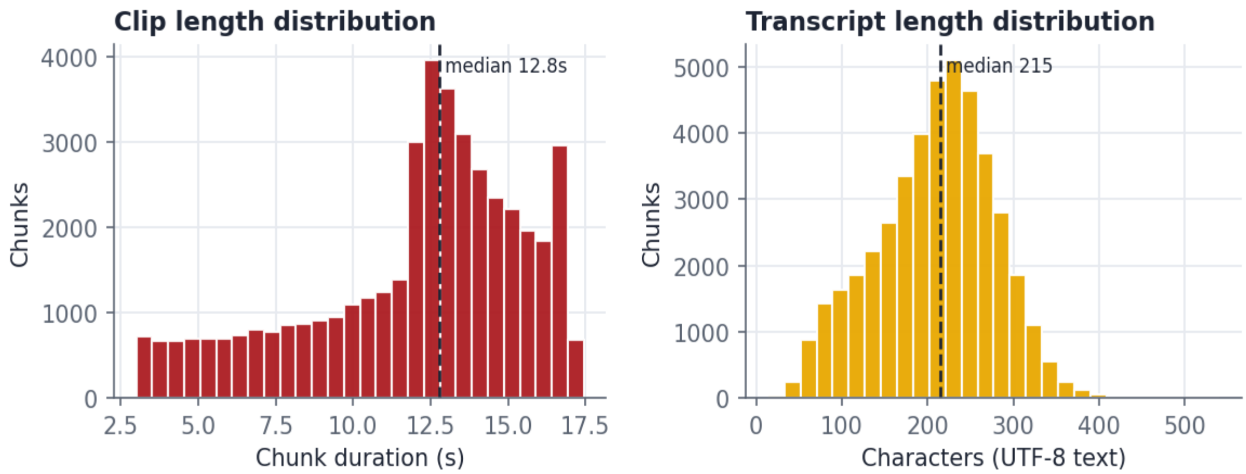

Figure 2 — Clip length (median 12.8 s) and transcript length (median 215 chars). Clips are capped around 17 s to fit the decoder's 1536-frame audio window.

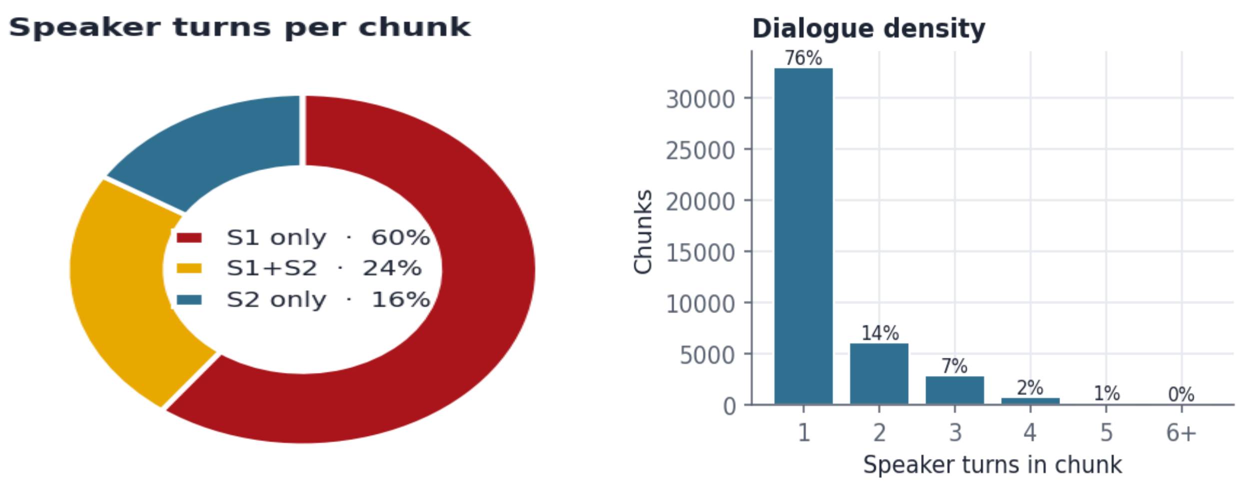

Figure 3 — Speaker composition and dialogue density. ~24% of clips contain a genuine [S1]↔[S2] exchange; the rest are single-speaker spans — useful for voice identity, but turn-taking is the scarcer, higher-value signal.

5 Training Setup

The original Dia release ships inference-only. The training harness adds: DAC-encoding of audio into the 9-channel target, construction of **delay-pattern** targets, a **channel-weighted cross-entropy** loss (4× weight on the first/coarsest codebook), and an 8-bit optimizer to fit the 1.6 B model on a single GPU.

Setting	Value
Method	Full fine-tune (all weights) from nari-labs/Dia-1.6B
Precision	bfloat16
Optimizer	bitsandbytes AdamW (8-bit)
Loss	Cross-entropy over 9 DAC channels, 4× weight on channel 0
Batch	2 × grad-accum 2 (smoke) → effective 4
Learning rate	1e-5 (smoke) · cosine schedule, 500 warmup steps
Smoke GPU	Modal L40S (~1 h, 1 epoch)
Target full run	A100-80GB, 43k chunks × 2 epochs (~24 h)
Tracking	Weights & Biases + TensorBoard
Checkpoints	ckpt_step501, ckpt_step1001, ckpt_epoch1 (on volume)

Note — plan vs. realized

The project config (`spanish_finetune.yaml`) originally specified a LoRA adapter on an A100. The realized harness instead does a **full** fine-tune with an 8-bit optimizer, validated at smoke scale on an L40S. The LoRA path remains a viable, cheaper option for the full run if memory or over-fitting to The Wild Project becomes a concern.

Evaluation plan

- **WER / CER** — transcribe generated audio with Whisper-large-v3, compare to the input text.
- **Speaker consistency** — verify [S1]/[S2] map to stable, distinct voices.
- **Naturalness (MOS)** — subjective rating of prosody, emotion and nonverbal cues.

6 Next Steps

- Launch the **full A100 run** (43k × 2 epochs) resuming from the smoke checkpoint.
- Add **source-balanced sampling** so The Wild Project does not dominate the learned voice.
- Run the **evaluation suite** on held-out prompts; compare base vs. fine-tuned.
- Mine more **multi-turn [S1]↔[S2]** material to strengthen turn-taking.
- Publish the dataset + checkpoints to the HuggingFace Hub.

D8 Labs · Dia-Spanish · AutoScientist Challenge (Adaption Labs) · Built on NARI Labs Dia & Descript Audio Codec